

web advertisement

1 Web Advertising

1.1 Concepts and Definitions

1. What is CPM pricing? How does it differ from pay-per-click (PPC) pricing?

Cost per mille impressions. It is priced based on the amount of times an add is viewed while ppc accounts for how often it is interacted with

2. What is the click-through rate (CTR) of an ad, and how is it computed?

CTR = number of clicks over the number of impressions

3. What is an Ad Rank (quality score) and why was it introduced?

It is used to decide what add to show to a user to increase the chance of interaction while also making google money

Bidxrelevance

Where bid is the amount the company will pay for a click while relevance is mainly calculated by the CTR but includes other factors

4. What is the difference between an on-line algorithm and an off-line algorithm?

An offline algorithm know what the chain of events will be and can optimally route add to the correct queries to maximes profit while an online one that is used in practice is one that must make the best decision at any given moment

5. Why does search advertising require an on-line algorithm?

It is a real time process

6. List the four structural constraints of the Adwords problem.

Each query can be matched to at most one advertiser

Advertiser budget cannot be exceeded

Decisions are irrevocable

It is an online algorithm

7. Define the competitive ratio of an on-line algorithm.

The competitive ratio is a scaler from 0 to 1 that when multiplied by the offline algorithms perfect output is equal to the on line algorithms output. It is the worst case fraction of offline optimal performance guaranteed by the online algorithm

8. In the Adwords formalization, define $N_t(q)$ and explain how it differs from $N(q)$.

$N_t(q)$ is the state of the graph at a given time and is thus the dynamic state while $N(q)$ is the initial static state.

1.2 Algorithms

1. Describe the greedy algorithm for the Adwords problem.

As a query arrives, assign the ad from the advertiser who has the largest bid.

2. Describe the MSVV (Balance) algorithm and explain how it differs from the greedy algorithm. Accounts for remaining budget with a scaled value and use it save advertiser budget to maximise spend across the day

3. Describe the Naive Balance algorithm and explain its decision rule.
Uses the add with the most remaining budget

1.3 Limitations and Trade-offs

1. List and briefly explain three limitations of the simple revenue-based ranking algorithm ($b \times \text{CTR}$).

Position bias: adds shown higher are artificially favoured and this may affect the observed CTR unfairly

Cold start: new adds have no historical ctr data

non-stationarity : CTR changes over time

Context dependance: CTR does not account for the full context of every scenario

2. Explain the exploration vs. exploitation trade-off in ad selection.

When new ads arrive we need to explore their usefulness to acquire a proper CTR that is representative while still ensuring that we push our best ads as much as possible.

3. Why can exhausting an advertiser's budget early be harmful, even if each individual allocation decision was locally optimal?

This advertiser may be valuable on future queries.

1.4 Analysis and Worked Problems

1. Bad scenario for greedy. Two advertisers A and B each have a budget of \$4. • A bids \$1 on query x only. • B bids \$1 on both query x and query y. The query stream is: x, x, x, x, y, y, y, y. a. Trace the greedy algorithm on this input. What is the total revenue? b. What is the optimal off-line allocation and its revenue? c. Compute the competitive ratio for this instance.

The worst case involves B being chosen the first 4 times exhausting there budget and leaving no more x's for advertiser a. The total revenue in this case is 4 while the optimal offline is 8 this gives us a c of 0.5

2. Naive Balance is not competitive. Two advertisers bid on query q: • A: bid = \$1, budget = \$110. • B: bid = \$10, budget = \$100. The query stream consists of 10 identical copies of q. a. Trace the Naive Balance algorithm on this input and compute the total revenue. 2 b. What is the optimal off-line revenue? c. Compute the competitive ratio for this instance. d. Explain why the competitive ratio of Naive Balance is 0 in general.

3. MSVV scaling behavior. Consider a single query q with two eligible advertisers: • A: bid = \$1, remaining budget fraction $f_A = 1.0$ (full budget). • B: bid = \$1, remaining budget fraction $f_B = 0.2$ (80\ a. Compute the MSVV scaled weight $\tilde{w}$ for each advertiser. (Use $e \approx 2.718$.) b. Which advertiser does MSVV select? Explain why this makes sense. The MSVV scaling formula is:

$$\tilde{w}_{q,a} = w_{q,a} \cdot (1 - e^{-f_a}).$$

$$\text{bid}_a(1 - e^{-f_a})$$

$$\text{For a } 1 \times (1 - e^{-1}) = 0.632$$

$$\text{For b } 1 \times (1 - e^{-0.2}) = 0.18$$

1. Revenue loss from shrinking neighborhoods. Consider three advertisers A, B, C with budgets $B_A = B_B = B_C = 2$ and the following bids (all \$1): • A bids on queries: p, q • B bids on queries: p, q, r • C bids on queries: r, s The query stream is: p, p, q, q, r, r, s, s. a. Trace the greedy algorithm. For each step, show $N_t(q_t)$, the assignment, and remaining budgets. b. What is the total revenue? c. What is the optimal off-line revenue? d. Identify the queries where revenue is lost because $N(q) \neq \emptyset$ but $N_t(q) = \emptyset$.

First p can go to either a or b

We will give it to a meaning a now has a budget of 1

1.5

True or False For each statement, indicate whether it is True or False and briefly justify your answer.

1. In CPM pricing, the advertiser pays only when a user clicks on the ad.

false

2. The Adwords problem is an on-line constrained allocation problem because queries arrive sequentially and decisions are irrevocable.

true

3. If an on-line algorithm has a competitive ratio of 0.75, it guarantees at least 75

true

4. The greedy algorithm for the Adwords problem has a competitive ratio of $1/2$ in the general (non-equal-bid) case.

false

5. The MSVV (Balance) algorithm achieves a worst-case competitive ratio of $1 - 1/e \approx 0.63$, and no deterministic on-line algorithm can do better.

true

6. The Naive Balance algorithm (assign to the advertiser with the largest remaining budget) has a competitive ratio of $1/2$.

false

7. In the Adwords bipartite graph formulation, the set of edges E changes over time as budgets are depleted.

false

8. An on-line algorithm that makes locally optimal decisions at each step is guaranteed to produce a globally optimal solution.

false

reccomendation

2 Recommendation Systems

2.1 Concepts and Definitions

1. What is the goal of a recommendation system?

To predict how much a user will like an item

2. What is the "long tail" and why is it relevant to recommendation systems?

There is a large number of niche items and those are the ones users need help finding and that is the power of recommendation systems

3. What is the utility matrix? Why is it typically sparse?

The utility matrix is the data we have linking users to items with their preferences. It is usually sparse because users only try a small portion of items

4. What is the difference between explicit and implicit ratings? Give one limitation of each.

Explicit ratings are direct ratings placed upon an item by the user while implicit ratings are derived based on the user's interaction with the item

5. What is the cold start problem in recommendation systems?

When you start the system you have no info to go off of

New items and new users have no history

6. List the three main approaches to recommendation and give a one sentence description of each.

Content based recommendation uses a feature vector based on details of the item to relate it to other items you have like before and recommend it to you

Collaborative filtering where use patterns of different items are matched with yours to give good recommendations

Latent factor based recommendations count on the fact that there is an unseen vector space that captures some kind of preference that can be used to pinpoint recommendations

2.2 Content-Based Recommendations

1. What is an item profile and how is it constructed? \

A feature representation of an item created from listed info about it

2. How is a user profile constructed in content-based recommendation?

Based on the history of interactions a map of the features a user prefers is recorded and turned into a profile

3. How does content-based recommendation use cosine similarity to make predictions?

They use it to analyse how closely an item's profile matches your user profile

4. List two advantages and two disadvantages of content-based recommendation.

Pros: it does not require data from other users, it can recommend new or unpopular items, and it can explain its recommendations

Cons: it requires good features and it struggles with new users and overspecialization

2.3 Collaborative Filtering

1. What is collaborative filtering? How does it differ from content-based recommendation?

Collaborative filtering relies on user patterns to recommend things while content based recommendation uses item description.

2. In collaborative filtering, why is using Jaccard similarity on the raw utility matrix problematic? It loses information about the users ratings of the product and oversimplifies simply to have they interacted. They may have interacted but it could have been a negative interaction

3. Why is it important to normalize ratings by the user's average before computing cosine similarity?

Because users rate with their own implicit bias so you have to standardize it against other people's style of rating

4. Explain the difference between user-user and item-item collaborative filtering.

User user matches patterns between similar users while item item matches use patterns between similar items

5. Write the formula for predicting a rating using similarity-weighted collaborative filtering (user-user variant). Define all terms.

Tomo

6. List three disadvantages of collaborative filtering.

cold start

sparsity

first-rater problem

popularity bias

difficulty helping users with unique tastes

2.4 Analysis and Worked Problems

1. Collaborative filtering with normalized ratings. Consider the following utility matrix (1–5 scale):

	M1	M2	M3	M4
Alice	5	3	1	
Bob	4	2	1	
Carol	2	5	4	

a. Compute the average rating for each user.

b. Compute the normalized utility matrix, where normalization is done relative to each user (subtracting each user's average). c. Using cosine similarity on the normalized ratings, which user is most similar to Alice? d. Using user-user collaborative filtering (with the most similar user), predict Alice's rating for M3.

Subtract each user's average rating from their individual rating to remove their bias

Use cosine similarity to compute the level of vector agreement

2. Content-based vs. collaborative filtering scenario. A new streaming platform has just launched with a catalog of 50,000 movies. It has 100 users who have each rated about 10 movies. Detailed metadata (genre, director, actors, plot keywords) is available for every movie.

a. Explain why collaborative filtering would perform poorly in this scenario.

There isn't much data to go off of as the utility matrix is very sparse

b. Explain why content-based recommendation is a better fit.

Since movie information is already available we can turn them into comparable feature vectors

c. As the platform grows to 10 million users, which approach becomes more attractive and why? Collaborative filtering can become more attractive because the system now has enough data to find strong similarity between users and items

2.5 True or False For each statement, indicate whether it is True or False and briefly justify your answer.

1. Content-based recommendation can recommend an item that no user has ever rated.

true

2. In collaborative filtering, treating missing entries in the utility matrix as 0 is equivalent to assuming the user disliked those items.

true

3. Jaccard similarity between two users captures whether they gave similar ratings to the same items.

false

4. Normalizing ratings by the user average before computing cosine similarity makes the result equivalent to Pearson correlation.

true

5. Collaborative filtering cannot suffer from overspecialization because it does not use item features.

false

6. Item-item collaborative filtering computes similarity between items based on their content features (genre, actors, etc.).

false

3 Recommendation Systems

2: Latent Factor Models and the Netflix Challenge

3.1 Concepts and Definitions

1. What was the Netflix Challenge, and what metric was used to evaluate submissions?

They wanted to improve their movie recommendation algorithm by 10% according to root mean squared error

2. What is the "surprisingly strong baseline" for the Netflix Challenge, and what lesson does it teach?

Rating is equal to the sum of the users average rating and the movies average rating over 2

3. What is the baseline predictor b_{xi} , and what are its components?

B_{xi} consists of the overall mean rating, the bias of user x , and the bias of item i

4. What is a latent factor? Why is it called "latent"?

A hidden dimension of preference across a utility matrix. It is not explicitly defined in the feature space but is learned over time

5. In latent factor models, what are the matrices U and V , and how are they used to predict ratings?

U is the user latent factor matrix and V is the item latent factor matrix. To predict the rating the users U is dot produced with the items V

6. Write the full latent factor model with biases. Define all terms.

$$R_{xi} = \mu + b_x + b_i + q_i \cdot p_x$$

7. What is overfitting in the context of matrix factorization, and how is it addressed?

Overfitting is when the model fits noise in training data and it is addressed using regularization

8. What is an interpolation weight w_{ij} in the improved collaborative filtering model, and how does it differ from similarity?

An interpolation weight is a learned coefficient that models how much rating item j helps predict item i .

3.2 Improving Predictions

1. The winning Netflix Challenge solution combined three types of predictive models. List them and briefly explain what each captures.

Bias models = models that capture user and item bias

Neighbourhood models = models that capture local trends among similar users and item

Latent factor models = models that capture hidden global preferences

2. List three limitations of neighborhood-based collaborative filtering that latent factor models address.

sparsity of the rating matrix

unreliable similarity estimates

inability to capture hidden preference structure

difficulty modeling deeper user-item interactions

3. Why did external data (IMDB metadata: actors, genres, directors) provide minimal improvement in the Netflix Challenge?

Latent factors already capture this info indirectly.

4. What are ensemble methods in recommendation, and why did the winning Netflix solution use them?

Because all models have weaknesses so making up for the weaknesses through combination of other models leads to a stronger and more consistent result

3.3 Analysis and Worked Problems

1. Baseline predictor computation. Assume that ratings range from 0 to 5. The overall mean rating is $\mu = 3.5$. Consider: • User Alice: rates 0.8 above average ($b_{\text{Alice}} = 0.8$). • User Bob: rates 0.5 below average ($b_{\text{Bob}} = -0.5$). • Movie X: rated 1.2 above average ($b_X = 1.2$). • Movie Y: rated 0.3 below average ($b_Y = -0.3$). a. Compute the baseline prediction for Alice on Movie X and for Bob on Movie Y. b. Alice actually rated Movie X a 4. What is the residual that the latent factor interaction $q_X \cdot p_{\text{Alice}}$ is expected to explain?

$$3.5 + 0.8 + 1.2 = 5.5$$

$$3.5 - 0.5 - 0.3 = 2.7$$

$$5.5 - 4 = 1.5$$

2. Latent factor prediction. Assume that ratings range from 0 to 5. Two users and three movies have the following 2-dimensional latent factor vectors: Users: $p_{\text{Alice}} = (1, 2)$, $p_{\text{Bob}} = (-1, 3)$.

Movies: $q_A = (2, 1)$, $q_B = (-1, 2)$, $q_C = (3, -1)$. Assume $\mu = 3$, all biases are 0 (for simplicity). a. Compute the predicted rating for each (user, movie) pair. b. Which movie should be recommended to Alice? To Bob?

$$2 + 2 = 4$$

$$-1 + 4 = 3$$

$$3 - 2 = 1$$

$$-2 + 3 = 1$$

$$1 + 6 = 7$$

$$-3 + -3 = -6$$

Alice a and bob b

3. RMSE computation. A recommendation system produces the following predictions: 8 User Movie Actual rating (r) Predicted rating ($\hat{r}$) Alice M1 5 4.2 Alice M2 2 2.8 Bob M1 4 3.5 Bob M3 1 2.0 a. Compute the RMSE. b. A baseline that always predicts the global average (3.0) is proposed. Compute its RMSE on the same data. Is the recommendation system better?

0.795

3.4 True or False For each statement, indicate whether it is True or False and briefly justify your answer.

1. The baseline predictor $b_{xi} = \mu + b_x + b_i$ captures user-item interactions.

false

2. In matrix factorization, increasing the number of latent factors k always improves prediction accuracy on unseen data.

false

3. The predicted rating $\hat{r}_{u,i} = U_u \cdot V_i$ can be computed in $O(k)$ time, where k is the number of latent factors.

true

4. Adding external metadata (actors, genres) to a latent factor model always substantially improves RMSE.

false

5. Regularization in matrix factorization prevents the model from fitting noise by penalizing large latent factor vectors.

true

6. In the Netflix Challenge, the winning solution used a single highly-tuned model rather than combining multiple models.

false

7. Latent factors in matrix factorization correspond to pre-defined categories such as "action" or "comedy."

false

8. Stochastic gradient descent is used to learn the parameters of the latent factor model because the objective function cannot be solved in closed form when the matrix has missing entries.

true

Association Rules

4 Association Rules

4.1 Concepts and Definitions

1. What is the market basket model? Define items, baskets, and the goal.

In the market basket model items are products to buy at the market and baskets represent market transactions. The goal is to find patterns of co-occurrence across items

2. Define the support count of an itemset I.

The support count of an item is the number of times that itemset appears as subset of a transaction

3. Define the support and confidence of an association rule $I \rightarrow j$.

The support of an association rule is the number of times both sides of the rule appear in a transaction and the confidence is the how likely the right side of the rule is to appear when the left side is present in other words it is the percentage of the time the right side occurs when the left side has occurred

4. What is the interest of an association rule, and why is confidence alone insufficient?

The interest adjusts for the baseline popularity to see if a rule explain a unique distinction or a capture of popular items.

5. State the a-priori property and explain why it is useful.

Every subset of a frequent itemset must also be frequent

6. Explain the monotonicity property of support: why is $\text{support}(Y) \leq \text{support}(X)$ whenever $X \subseteq Y$?

Because every basket containing the larger itemset Y must also contain the smaller itemset X. Adding more items can only keep support the same or reduce it

7. What is a maximal frequent itemset? What information does it lose compared to storing all frequent itemsets?

A maximal frequent itemset is an itemset with no frequent supersets. It loses direct information about the support count but guarantees that every subset within it must at least be frequent

8. What is a closed frequent itemset? How does it differ from a maximal frequent itemset?

A closed frequent itemset is an itemset with no superset that has the same support. This preserves information about the support.

4.2 Algorithms

1. Describe the A-priori algorithm. What does each pass accomplish?

Because every basket containing the larger itemset YYY must also contain the smaller itemset XXX. Adding more items can only keep support the same or reduce it

2. Why is counting pairs the memory bottleneck of A-priori? Compare the triangular matrix and triples methods.

There are about $n^2/2$ possible pairs in a set of n items so pair counting is very large

A triangular matrix can store all pair counts in half of a matrix and is better when about a 1/3 of the possible pairs exist.

The triples method uses dictionaries to store observed pairs and it better when the matrix is to sparse

3. Describe the PCY (Park-Chen-Yu) algorithm and explain how it reduces memory usage in Pass 2.

Use a hashmap to hash pairs to buckets and store a count there. We can then use this in step 2 to prune the pairs that are not frequent

4. What is the multistage improvement to PCY? What does it gain and what does it cost?

Add another hash step to the process and hash once more the results of the first hash to decrease false positives it adds an extra pass and extra memory though

5. What is the multihash variant? What is its risk compared to multistage?

Use multiple hashes in the first pass to save on time but increase on memory

6. Describe the SON algorithm and explain why it has no false negatives, and why it has false positives.

Split the dataset into chunks and compute locally adjusted frequent itemsets. If an itemset is frequent it will appear as locally frequent in at least one chunk. If the itemset happens to be only locally frequent in one chunk and nowhere else then you get a false positive.

7. What are the trade-offs of using random sampling to find frequent itemsets?

4.3 Applications

1. Give three applications of the market basket model beyond supermarket transactions.

plagiarism detection

related concept / word co-occurrence in text

product recommendation such as "people who bought X also bought Y"

4.4 Analysis and Worked Problems

1. Support, confidence, and interest. Consider 1000 transactions with the following contingency table: Coffee Not Coffee Total Tea 150 50 200 Not Tea 650 150 800 Total 800 200 1000 a. Compute $\text{support}(\{T \rightarrow \{Coffee\}\})$. b. Compute $\text{confidence}(\{T \rightarrow \{Coffee\}\})$. c. Compute $\text{interest}(\{T \rightarrow \{Coffee\}\})$. d. Is this rule interesting? Explain.

2. A-priori algorithm trace. Consider the following transactions with minimum support threshold $s = 3$: tid items 1 {m, c, b} 2 {m, p, j} 3 {m, c, b, n} 4 {c, j} 5 {m, p, b} 6 {m, c, b, j} 7 {c, b, j} 8 {b, c} a. Find all frequent singletons (L1). b. Find all frequent pairs (L2). c. Generate candidates C3 from L2 and determine L3. d. From the frequent itemset {m, c, b}, generate all association rules and compute their confidence. Which rules have confidence ≥ 0.75 ?

3. PCY hash filtering. Suppose in Pass 1 of PCY, there are 6 frequent items: {a, b, c, d, e, f}. The hash function maps pairs to 5 buckets as follows: Pair Bucket (a,b) 1 (a,c) 2 (a,d) 3 (b,c) 1 (b,d) 4 (c,d) 2 (a,e) 5 (b,e) 3 (other) 5 After Pass 1, bucket counts are: bucket 1: 5, bucket 2: 2, bucket 3: 4, bucket 4: 1, bucket 5: 3. The minimum support is 3. a. Which buckets are frequent? Construct the bitmap. b. Which of the listed pairs are candidate pairs in Pass 2? c. Pair (c, d) has true support 4. Is it a candidate? Explain the consequence.

4. Memory comparison. A dataset has $n = 20,000$ distinct items. Each counter requires 4 bytes.

a. How much memory is needed for a triangular matrix to count all pairs? b. Suppose only 5 c. Which method is more memory-efficient for this dataset?

5. Comprehensive A-priori analysis. Consider the following 8 transactions with minimum support threshold $s = 3$: tid items 1 {a, b, c, d} 2 {a, b, c, d} 3 {a, b, c, e} 4 {a, c, d, e} 5 {b, c, d, e} 6 {a,

b, c\ 7 \d, e\ 8 \a, b\ a. Run the A-priori algorithm to find all frequent itemsets. Show L1, L2, C3, L3, and C4. b. For each frequent itemset of size 3, generate all association rules with a 2-item left-hand side. Compute the confidence and interest of each rule. Which rule is the most interesting? c. Find all maximal frequent itemsets. d. Find all closed frequent itemsets (with their support counts). Identify which frequent itemsets are not closed and explain why.

6. Maximal and closed frequent itemsets. The following is the complete list of frequent itemsets (with support counts) for a dataset of 12 transactions with minimum support threshold $s = 4$:
 Size Frequent itemsets
 1 \a\ :9, \b\ :7, \c\ :7, \d\ :6, \e\ :5
 2 \a,b\ :7, \a,c\ :5, \b,c\ :7, \b,d\ :4, \c,d\ :4, \d,e\ :5
 3 \a,b,c\ :5, \b,c,d\ :4
 a. Find all maximal frequent itemsets. b. Find all closed frequent itemsets with their support counts. c. Which frequent itemsets are not closed? For each, name the superset that proves it is not closed. d. Verify that the support of every non-closed itemset can be recovered from the closed frequent itemsets alone.

7. Reconstructing frequent itemsets from maximal frequent itemsets. The maximal frequent itemsets for a dataset are: {a, b, c, d}, {a, c, e}, {b, e, f}
 a. List all frequent itemsets implied by these maximal itemsets. Organize them by size. b. How many frequent itemsets are there in total? c. A student claims that {a, b, e} is frequent. Is this necessarily true? Explain. d. Can you determine the support count of {a, b}? Explain what information is lost when only maximal frequent itemsets are stored.

4.5 True or False

1. Confidence of a rule $I \rightarrow J$ is symmetric: $\text{confidence}(I \rightarrow J) = \text{confidence}(J \rightarrow I)$.

false

2. If an itemset {A, B, C} is not frequent, then {A, B, C, D} is also not frequent.

true

3. PCY can produce false negatives: a truly frequent pair might not be counted in Pass 2.

false

4. The SON algorithm can produce false negatives: a globally frequent itemset might not appear as a candidate.

false

5. A maximal frequent itemset is always a closed frequent itemset.

true

6. In the multihash variant of PCY, using more hash functions in Pass 1 always reduces the number of candidate pairs.

false

7. Random sampling for frequent itemsets can eliminate false positives with an additional verification pass, but cannot eliminate false negatives.

true

Graph Analysis

5 Network / Graph Analysis

5.1 Concepts and Definitions

1. What is locality in a social network, and how does it differ from a random graph?

Locality describes the likelihood in a graph the neighbours of a node are likely to also be neighbours of each other

2. What is edge betweenness?

Edge betweenness is the number of pairs of nodes whose shortest path to each other contains this edge

3. Describe the Girvan-Newman (GN) algorithm for finding communities.

Compute the edge betweenness and then remove the edge with the highest betweenness continue this until your desired stopping point

4. In the GN algorithm, how is betweenness computed for a single source node?

Build the shortest bfs tree from that source to all other nodes. Count how many shortest paths reach each node. Distribute credits back from the leaves towards the source

5. What is modularity? Write the formula and define all terms.

$$Q(G,S) = \frac{1}{2m} \sum_{s \in S} \sum_{i \in s} \sum_{j \in s} (A_{ij} - \frac{k_i k_j}{2m})$$

where:

- G = graph
- S = partition of nodes into communities
- m = total number of edges
- $A_{ij} = 1$ if edge (i,j) exists, else 0
- k_i = degree of node i

The term A_{ij} is the observed connection, and $\frac{k_i k_j}{2m}$ is the expected connection under the null model.

6. What is the null model used in modularity, and why is it degree-preserving?

The null model is a random graph with the same degree distribution as the original

7. Interpret the following modularity values: $Q \approx 0$, $Q \geq 0.3$, $0.3 \leq Q \leq 0.7$, $Q > 0.7$.

Q is low between 0 and 0.3 meaningful above 0.3 and strong closer to 0.7 beyond 0.7 it becomes unusual suggesting the graph is artificial

8. How is modularity used to determine the optimal number of communities?

Compute the modularity of different partition possibilities and choose the highest

9. List three types of relationships that can exist in a social network.

Binary

Categorical

numerical

10. List four applications of network analysis.

community detection

recommendation systems

information diffusion / rumor tracking

link prediction

influence analysis
anomaly or fraud detection

5.2 Analysis and Worked Problems

1. Locality computation. Consider a graph with 7 nodes and 9 edges: A --- B --- D --- E | / | / | / | / | C G --- F Edges: (A,B), (A,C), (B,C), (B,D), (D,E), (D,G), (D,F), (G,F), (E,F). a. How many triangles are in this graph? List them. 15 b. For a random graph with 7 nodes and 9 edges, what is the probability that an edge exists between two arbitrary nodes? c. Given that edges (X,Y) and (X,Z) exist, what is the probability that (Y,Z) exists in the random graph?

2. Betweenness computation. Consider the following graph: B / \ A D --- E \ / C Edges: (A,B), (A,C), (B,D), (C,D), (D,E). a. Compute the betweenness of each edge. b. Which edge would the Girvan-Newman algorithm remove first? c. After removing that edge, what are the resulting connected components? 3.

Modularity computation. Consider a graph with 8 nodes and 10 edges, partitioned into two communities $S = \{s_1, s_2\}$: $s_1 = \{A, B, C, D\}$ with internal edges: (A,B), (A,C), (A,D), (B,C), (B,D), (C,D). That is 6 edges (a complete subgraph). $s_2 = \{E, F, G, H\}$ with internal edges: (E,F), (E,G), (F,H). That is 3 edges. Inter-community edge: (D,E). That is 1 edge. Total: $m = 10$ edges. a. For community s_1 , compute the number of internal edges and the expected number under the null model. b. Compute the modularity Q of this partition. c. Is this a good partition? Justify using the modularity interpretation guidelines. 16

4. Evaluating a bad partition. Using the same graph from Q13 (8 nodes, 10 edges), consider an alternative partition: $s'_1 = \{A, B, E, F\}$ and $s'_2 = \{C, D, G, H\}$. a. How many internal edges does each community have? b. Without computing the full modularity, explain why this partition is likely to have a lower Q than the original.

5.3 True or False For each statement, indicate whether it is True or False and briefly justify your answer.

1. In a social network, if A is friends with B and A is friends with C, then B and C are always friends.

false

2. Edge betweenness measures the degree (number of neighbors) of the nodes at each end of the edge.

false

3. In the Girvan-Newman algorithm, betweenness must be recomputed after each edge removal.

true

4. A modularity value of $Q = 0$ means the graph has no edges.

false

5. Maximizing modularity always finds the unique correct community structure.

false

6. If edge (D,E) has the highest betweenness in the graph, it necessarily connects two different communities.

false

7. Modularity can be negative.'

trueok